

Question 1:

* COUNT () Function

```
SELECT COUNT('id') AS employees
FROM employees2;
```

employees
10

Question 2:

* SUM () Function

```
SELECT SUM(salary) AS total_salary
FROM employees2
WHERE department = 'IT';
```

total_salary
220000

Question 3:

~~* AVG(salary) AS average_salary~~

* AVG () Function

```
SELECT AVG(salary) AS average_salary
FROM employees2
WHERE department = 'HR';
```

average_salary
49500

Question 4:

* MIN () and MAX () Function

```
SELECT
    MIN(salary) AS lowest_salary,
    MAX(salary) AS highest_salary,
FROM employees2;
```

lowest_salary	highest_salary
48000	62000

Question 5:

* GROUP BY Statement

```
SELECT department
    SUM(salary) AS total_salary
FROM employees2
GROUP BY city;
```

department	total_salary
IT	220000
HR	49000
Finance	119000
Marketing	105000

Question 6:

* GROUP BY and COUNT

```
SELECT city,
       COUNT('employees') AS total_employees
FROM employees2
GROUP BY city;
```

City	total_employees
New York	2
Chicago	3
Los Angeles	2
San Francisco	2
Houston	1

Question 7:

* GROUP BY and ORDER BY

```
SELECT city,
       COUNT('employee') AS total_employee
FROM employees2
GROUP BY city
ORDER BY total_employees DESC;
```

City	total_employees
Chicago	3
New York	2
San Francisco	2
Los Angeles	2
Houston	1

Question 8:

* Having Clause

```
SELECT department
       SUM(salary) AS total_salary
FROM employees2
GROUP BY department
HAVING SUM(salary) > 100000
```

department	total_salary
IT	220000
Finance	119000
Marketing	105000

Question 9:

* Combining GROUP BY, HAVING and ORDER BY

```
SELECT city,
       COUNT('employees') AS total_employees
FROM employees2
GROUP BY city
HAVING COUNT('employees') > 1
ORDER BY total_employees DESC;
```

City	total_employees
Chicago	3
New York	2
Los Angeles	2
San Francisco	2

Question 10:

* Combining Aggregate functions

```
SELECT department
       AVG(salary) AS avg_salary
FROM employees2
GROUP BY department
ORDER BY avg_salary DESC
LIMIT 1;
```

department	avg_salary
Finance	59500